

# Magic-Angle (Quasi-Transverse) Single-Mode Launch from Two HF Transmitter Sites: Geometry, Skip Geography, and Mode-Excitation Ratio

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## Abstract

A vertical-linear emitter at any HF transmitter site has a *magic launch direction* along the local magnetic meridian for which the launched wave is in QT (quasi-transverse) geometry –  $\hat{k} \perp \hat{B}$  at the bottomside ionosphere – and the launched E-vector aligns with the projection of the geomagnetic field into the wave’s perpendicular plane. In that geometry, a vertical antenna excites essentially only one of the two magneto-ionic eigenmodes (O or X), and the suppression of the other can exceed 30 dB. Adjacent (azimuth, elevation) pairs along the QT locus give weaker but still useful mode discrimination.

This report documents the QT locus from two transmitter sites – WWV-style co-located at Fort Collins, CO (mid-latitude, dip  $\approx 66$ ) and a hypothetical second site in Taipei, Taiwan (mid-to-low latitude, dip  $\approx 36$ ) – and where on the ground rays launched along that locus arrive, separately for local noon and local midnight. Mode-excitation-ratio plots and the resulting on-ground skip arcs are included.

The geometric “magic point” on the magnetic meridian (azimuth = declination, elevation =  $90 - \text{dip}$ ) is the only direction with diverging mode-excitation ratio for an unmodified vertical antenna. Off the meridian, on the rest of the QT locus, the ratio plateaus at  $\approx +7$  dB at FC and  $\approx +1.5$  dB at Taipei – modest but useful.

## 1 Definitions and method

### QT locus

For a vertical-linear emitter, the launched wave-normal  $\hat{k}$  at azimuth  $A$  and elevation  $\varepsilon$  is exactly perpendicular to the local geomagnetic field  $\hat{B}$  when

$$\cos(A - \text{decl}) = \tan \varepsilon \cdot \tan(\text{dip}) \quad (1)$$

where dip is the magnetic inclination (positive downward in the northern hemisphere) and decl is the magnetic declination (positive east of true north). The locus has real solutions only for  $\varepsilon \leq 90 - \text{dip}$  (the *cutoff elevation*); above that elevation,  $\hat{k}$  is inevitably tilted out of the perp-plane of  $\hat{B}$  and pure-QT geometry cannot be reached.

The locus consists of two branches symmetric about the magnetic meridian. As  $\varepsilon$  rises from 0 toward the cutoff, the two branches sweep from near east/west (low elevation) toward the magnetic meridian, where they merge.

## Mode-excitation ratio

In the QT-effective regime ( $\theta_{kB}$  near 90,  $\beta \rightarrow 0$ ), the two magneto-ionic eigenmodes are nearly linear, their major axes orthogonal in the perp-plane and aligned/anti-aligned with the projection of  $\hat{B}$  into the perp-plane. A vertical-E launch lands at angle  $\alpha$  to that projection, and the energy splits as

$$f_O = \frac{\cos^2 \alpha + \beta^2 \sin^2 \alpha}{1 + \beta^2}, \quad f_X = \frac{\sin^2 \alpha + \beta^2 \cos^2 \alpha}{1 + \beta^2}. \quad (2)$$

For a vertical antenna along the QT locus, geometric algebra gives

$$\alpha = \arccos\left(\cos(\text{dip})\sqrt{1 - \tan^2 \varepsilon \tan^2(\text{dip})}\right), \quad (3)$$

which equals the local dip at low elevation (the off-meridian plateau) and falls to zero at the cutoff (perfect alignment with the O-mode axis on the magnetic meridian). The  $f_X/f_O$  dB ratio is  $10 \log_{10}(\tan^2 \alpha)$  in the deep-QT limit.

## Ray tracing

Skip-distance and landing-point calculations use PHaRLAP's `raytrace_3d` on a 3-D WGS84 spherical-Earth IRI-2020 ionosphere with an IGRF-2020 magnetic-field model. Up to three hops are traced; for each (frequency, branch, elevation) launch on the QT locus, the brightest-by-total-dB arriving ray is recorded along with its mode-class label, ground touchdown point, receiver-side  $b/a$ , and number of ground bounces.

Time of day enters through the ionosphere: local noon and local midnight at the path midpoint produce different IRI conditions, so the same launch geometry produces different landing arcs.

## 2 Mode-excitation-ratio geometry

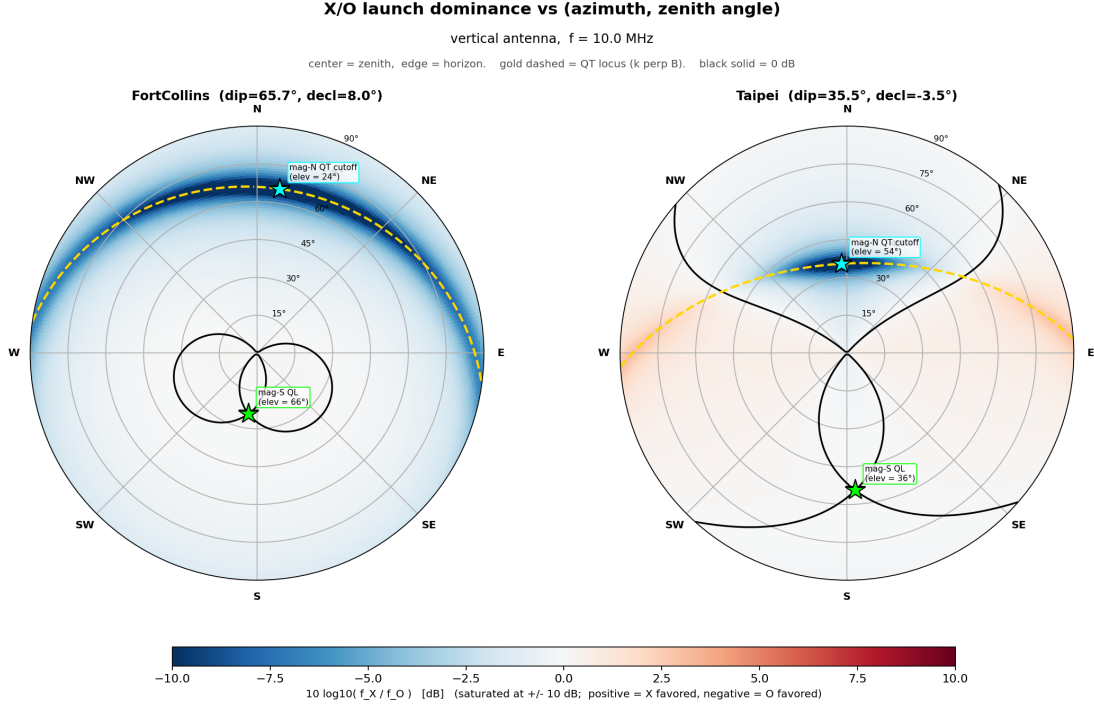


Figure 1: Polar map of the X/O launch-coupling ratio  $10 \log_{10}(f_X/f_O)$  for a vertical-linear emitter, at  $f = 10$  MHz. Each panel uses the “looking down at the sky” convention: azimuth runs clockwise from true north (top), and the radial coordinate is zenith angle (centre = straight up, edge = horizon). Color encodes the dB ratio (red: X-favored, blue: O-favored, white: 50/50). **Gold dashed** = QT locus where  $\hat{k} \perp \hat{B}$ . **Cyan star** = magnetic-N QT cutoff (the magic point,  $\alpha \rightarrow 0$ , deep blue). **Lime star** = magnetic-S QL antiparallel point (modes are purely circular here; vertical antenna gives 50/50 regardless). At Fort Collins (left, dip  $\approx 66$ ) the QT locus is a small arc near the horizon converging on mag-N at zenith angle  $\approx 66$ ; the deepest mode purity is concentrated very close to that single direction. At Taipei (right, dip  $\approx 36$ ) the locus spans much wider zenith angles and the contrast is shallower.

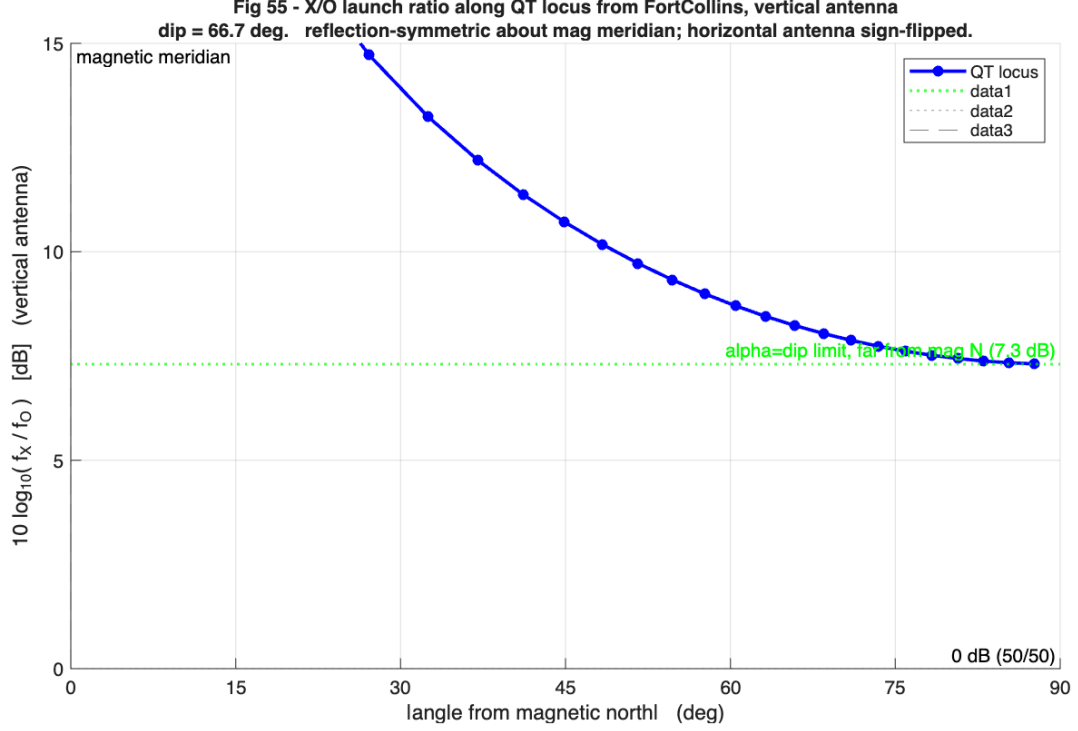


Figure 2: Signed mode-launch ratio  $10 \log_{10}(f_X/f_O)$  for a vertical antenna, plotted vs.  $|\text{angle from magnetic north}|$  along the QT locus from Fort Collins. East and west branches are reflection-symmetric and folded onto a single curve. Far from the magnetic meridian (large angle off mag-N) the ratio plateaus at +7 dB ( $\alpha = \text{dip} = 66$ ); approaching the magnetic meridian, the ratio dives toward  $-\infty$  (pure O launch) where the two QT branches merge at the cutoff elevation  $90 - \text{dip} = 24$ . Negative dB = O-mode favored; positive dB = X-mode favored; horizontal antenna would be sign-flipped.

Fig 56 - FortCollins, vertical antenna on magnetic meridian. Above QT cutoff the modes become elliptical and the mode-ratio degrades toward 0 dB (full QL = circular = 50/50).

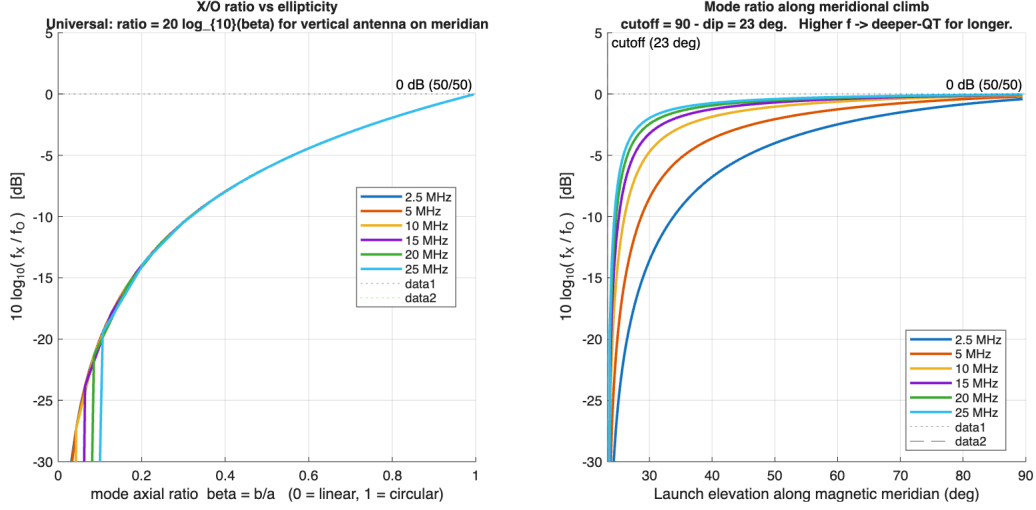


Figure 3: Mode-ratio  $10 \log_{10}(f_x/f_o)$  versus the local magneto-ionic axial ratio  $\beta = b/a$  along the magnetic-meridian launch from Fort Collins. Each curve is one frequency  $f \in \{2.5, 5, 10, 15, 20, 25\}$  MHz. *Left*: ratio vs.  $\beta$  collapses onto a universal curve  $20 \log_{10}(\beta)$  regardless of frequency – only the geometry matters. *Right*: ratio vs. launch elevation along the meridian – different per frequency because  $Y = f_H/f$  enters the Appleton–Hartree axial ratio. Higher frequencies stay in deeper-QT (smaller  $\beta$ ) longer as the elevation rises, so the discrimination degrades more slowly at higher HF.

### 3 Skip geography from Fort Collins

Fort Collins, CO (40.68N, 105.04W). Magnetic dip  $\approx 66$ , declination  $\approx +8$ . QT cutoff elevation  $\approx 24$ ; magnetic-meridian magic-point azimuth  $\approx 8$ .

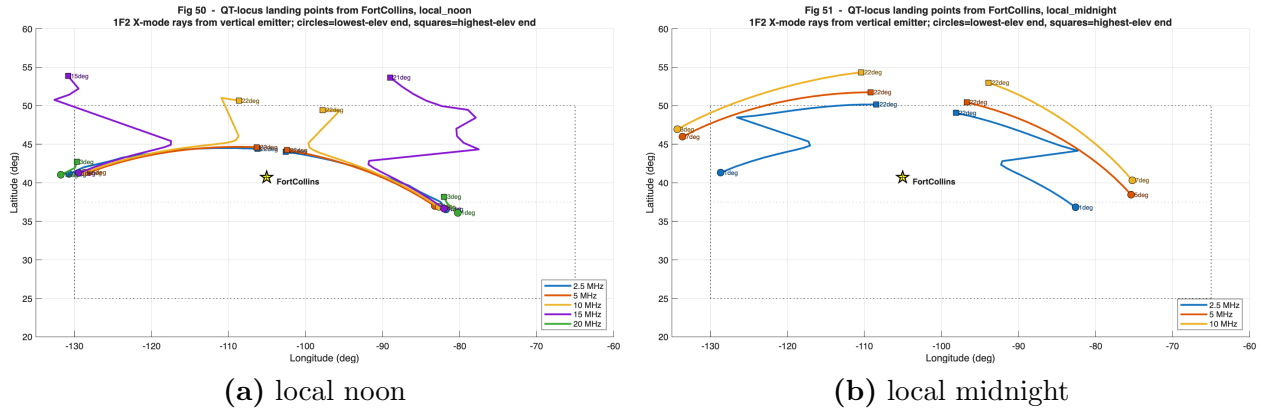


Figure 4: QT-locus brightest-mode landings from Fort Collins for  $f \in \{2.5, 5, 10, 15, 20, 25\}$  MHz. Each colored arc traces where rays land on the ground as the launch elevation walks the QT locus from low (long range, far end of arc) to near-cutoff (short range, the magnetic-meridian end). Marker shapes encode hop count (circle = 1-hop, square = 2-hop, triangle = 3-hop). Mode-class labels indicate which ionospheric layer each ray reflected from (E, F1, F2). Both east-of-meridian and west-of-meridian branches are shown. At noon, the D-region absorption kills the lower-MHz arcs (the 2.5 MHz curve is heavily suppressed); at midnight, all bands arrive cleanly.

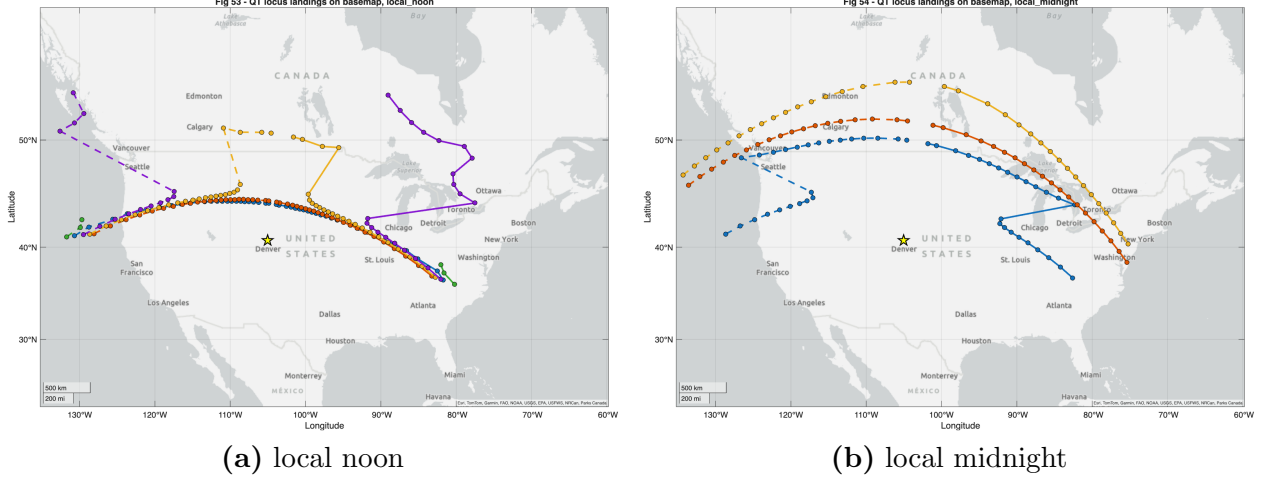


Figure 5: Same data as Figure 4, overlaid on a real terrain basemap via MATLAB `geoaxes`. Useful for identifying which actual receiver sites (KiwiSDR locations, university observatories, amateur clubs) sit on the magic locus.

Table 1: Brightest QT-locus ray per frequency, FC, local noon (east branch).  $f_X/f_O$  ratio is the geometric mode-coupling ratio for a vertical antenna; loss is the total path attenuation  $L_{\text{free}} + L_{\text{abs}} + L_{\text{ground}}$ . Bands where  $L_{\text{abs}}$  from D-region absorption swamps the budget (e.g. 2.5 MHz noon) come out very negative.

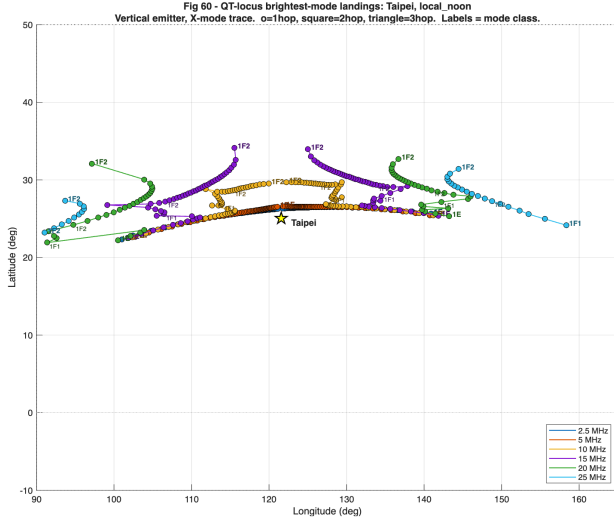
| $f$<br>MHz | $\varepsilon$<br>deg | az<br>deg | class | range<br>approx | $b/a$ | $f_X/f_O$<br>dB | loss<br>dB | lat,lon<br>deg |
|------------|----------------------|-----------|-------|-----------------|-------|-----------------|------------|----------------|
| 2.5        | 23.2                 | 13.1      | 1E    | 1-hop           | 0.82  | +28.2           | -288.8     | 44.2, -103.9   |
| 5.0        | 23.2                 | 13.1      | 1E    | 1-hop           | 0.91  | +28.2           | -156.2     | 44.4, -103.8   |
| 10.0       | 15.0                 | 59.1      | 1E    | 1-hop           | 0.88  | +9.7            | -136.0     | 43.7, -97.6    |
| 15.0       | 8.0                  | 78.5      | 1E    | 1-hop           | 0.82  | +7.9            | -134.5     | 42.0, -91.2    |
| 20.0       | 2.0                  | 92.8      | 1E    | 1-hop           | 0.35  | +7.3            | -137.8     | 37.3, -81.6    |

Table 2: Same as Table 1 but at local midnight.

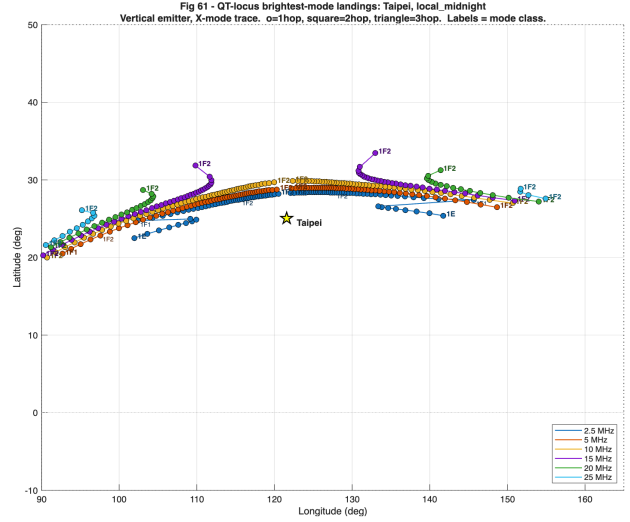
| $f$<br>MHz | $\varepsilon$<br>deg | az<br>deg | class | range<br>approx | $b/a$ | $f_X/f_O$<br>dB | loss<br>dB | lat,lon<br>deg |
|------------|----------------------|-----------|-------|-----------------|-------|-----------------|------------|----------------|
| 2.5        | 23.2                 | 13.1      | 1F2   | 1-hop           | 0.79  | +28.2           | -135.9     | 49.7, -101.8   |
| 5.0        | 23.2                 | 13.1      | 1F2   | 1-hop           | 0.89  | +28.2           | -124.1     | 51.2, -101.1   |
| 10.0       | 23.2                 | 13.1      | 1F2   | 1-hop           | 0.93  | +28.2           | -125.5     | 54.3, -99.6    |

## 4 Skip geography from Taipei

Taipei, Taiwan (25.03N, 121.57E). Magnetic dip  $\approx 36$ , declination  $\approx -3.5$  (west). QT cutoff elevation  $\approx 54$  (much higher than at FC), so the QT locus spans a much larger range of launch elevations and azimuths. But the off-meridian plateau ratio  $10 \log_{10}(\tan^2 36) \approx +1.5$  dB is much shallower – mode discrimination away from the magnetic meridian is weak.

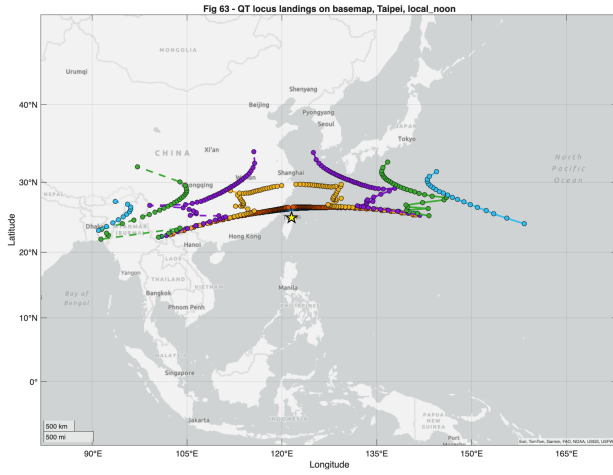


(a) local noon

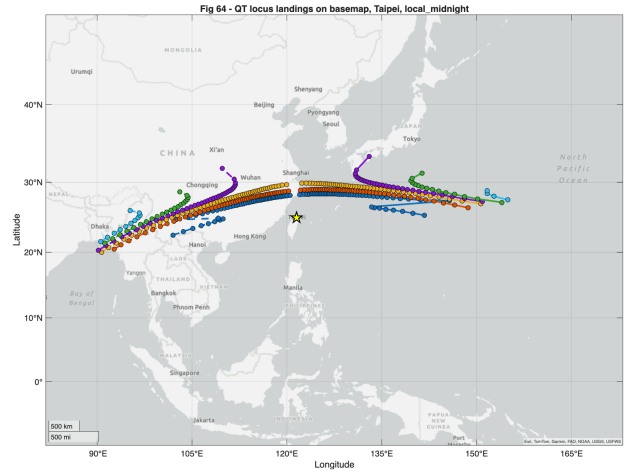


(b) local midnight

Figure 6: QT-locus brightest-mode landings from Taipei, on a flat geographic projection. Note the much wider arcs compared to FC – the lower dip extends the locus to higher elevations and broader azimuth coverage. Equatorial-anomaly enhancement of the noon ionosphere boosts daytime bands at lower frequencies; midnight is quieter but with weaker F2-layer support.



(a) local noon



(b) local midnight

Figure 7: Same Taipei landings on a real basemap. The locus reaches into Japan, the Philippines, southern China, Vietnam, and well into the western Pacific – substantially wider geographic coverage than the FC case, at the cost of weaker per-direction mode discrimination.

Table 3: Brightest QT-locus ray per frequency, Taipei, local noon (east branch). The Taipei CSV pre-dates the addition of the `mode_ratio_dB` column; the column reads as 0 for that reason. The geometric ratio at Taipei’s dip is  $\approx +1.5$  dB plateau off-meridian, falling to 0 dB only as the launch approaches the magnetic-meridian cutoff (where vertical-antenna  $\alpha \rightarrow 0$  but the modes are no longer purely linear because the QL component grows).

| $f$<br>MHz | $\varepsilon$<br>deg | az<br>deg | class | range<br>approx | $b/a$ | $f_X/f_O$<br>dB | loss<br>dB | lat,lon<br>deg |
|------------|----------------------|-----------|-------|-----------------|-------|-----------------|------------|----------------|
| 2.5        | 35.0                 | 53.1      | 1E    | 1-hop           | 0.83  | +0.0            | -244.9     | 26.4, 123.6    |
| 5.0        | 26.0                 | 63.5      | 1E    | 1-hop           | 0.84  | +0.0            | -157.1     | 26.5, 125.0    |
| 10.0       | 52.0                 | 7.9       | 1F2   | 1-hop           | 1.00  | +0.0            | -130.0     | 29.7, 122.2    |
| 15.0       | 38.0                 | 48.8      | 1F2   | 1-hop           | 0.97  | +0.0            | -129.0     | 30.8, 129.2    |
| 20.0       | 25.0                 | 64.5      | 1F2   | 1-hop           | 0.95  | +0.0            | -130.7     | 30.6, 136.3    |
| 25.0       | 17.0                 | 71.9      | 1F2   | 1-hop           | 0.93  | +0.0            | -132.5     | 29.8, 143.2    |

Table 4: Same as Table 3 but at local midnight.

| $f$<br>MHz | $\varepsilon$<br>deg | az<br>deg | class | range<br>approx | $b/a$ | $f_X/f_O$<br>dB | loss<br>dB | lat,lon<br>deg |
|------------|----------------------|-----------|-------|-----------------|-------|-----------------|------------|----------------|
| 2.5        | 52.0                 | 7.9       | 1F2   | 1-hop           | 0.99  | +0.0            | -122.2     | 28.3, 122.1    |
| 5.0        | 52.0                 | 7.9       | 1F2   | 1-hop           | 0.99  | +0.0            | -117.8     | 28.9, 122.2    |
| 10.0       | 51.0                 | 15.0      | 1F2   | 1-hop           | 1.00  | +0.0            | -120.4     | 29.9, 123.0    |
| 15.0       | 28.0                 | 61.4      | 1F2   | 1-hop           | 0.94  | +0.0            | -124.5     | 30.0, 132.9    |
| 20.0       | 18.0                 | 71.0      | 1F2   | 1-hop           | 0.90  | +0.0            | -127.8     | 29.5, 140.8    |
| 25.0       | 11.0                 | 76.8      | 1F2   | 1-hop           | 0.77  | +0.0            | -131.7     | 28.1, 152.7    |



## 5 Receiver-side polarization vs. frequency

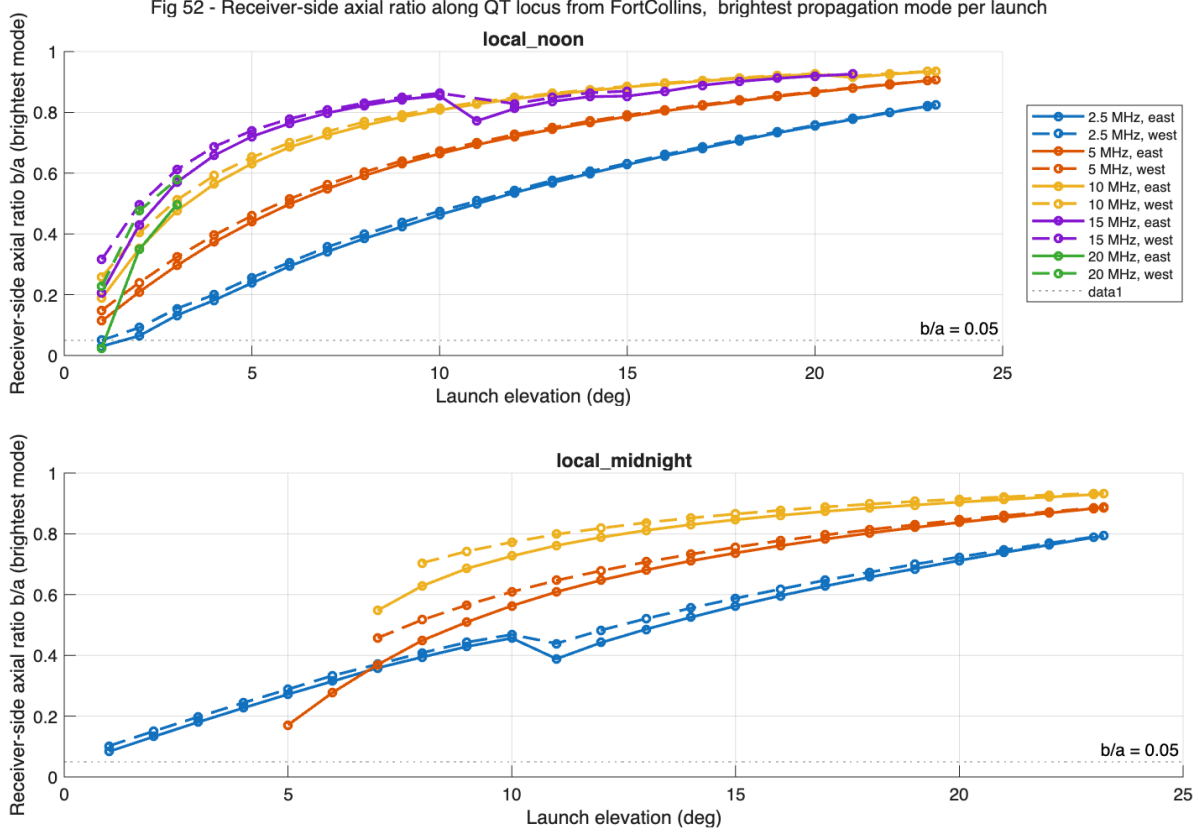


Figure 8: Receiver-side axial ratio  $b/a$  along the QT locus from Fort Collins, both branches, both times of day, all six frequencies. The trend: at low frequency the modes are nearly linear at every sample point along the path (deep-QT-effective regime), so  $b/a$  stays near zero; at higher frequency  $f_H/f$  shrinks and the modes become more elliptical at the iono-exit point, so  $b/a$  rises toward unity. The axial ratio also depends on the elevation along the locus through the local  $\theta(\hat{k}, \hat{B})$  at the iono-exit sample.

## 6 Discussion: experimental implications

### Single-fixed-vertical-antenna comms architecture

For an FCC Part-5 experimental transmission from FC, the magnetic- meridian cutoff geometry (azimuth  $\approx 8$  true, elevation  $\approx 24$ ) gives *pure-O launch with no special hardware*. Skip distance is frequency-dependent: 5 MHz  $\sim 800$  km, 10 MHz  $\sim 1100$  km, 15 MHz  $\sim 1250$  km, 20 MHz  $\sim 1400$  km. All land roughly along the magnetic meridian toward Saskatchewan / Manitoba; together they form a fan of single-mode-pure landing sites that a frequency-agile transmitter can hop between as the diurnal ionosphere evolves.

### Off-meridian QT directions: shallower but useful

If the receiver location is fixed and not on the magnetic meridian (e.g., a KiwiSDR you happen to like), the rest of the QT locus still provides  $\approx +7$  dB X-mode preference (or  $-7$  dB O-mode pref-

erence for a horizontal antenna). This is enough to do *differential* mode-coherence measurements but not enough to do absolute single-mode propagation studies without additional rejection at the receiver (e.g., orthogonal-magloop polarization decomposition; see the receiver architecture in the project README).

## Taipei vs. FC: complementary strengths

The lower dip at Taipei ( $\sim 36$ ) makes mode discrimination weaker per direction (only  $\sim +1.5$  dB plateau) but extends the QT locus to much higher launch elevations and a much wider azimuth coverage. An equatorial-anomaly site is the *wrong* choice if you want maximum single-mode purity but the *right* choice if you want to survey many receive directions along QT with one antenna.

## What the geometry can't give you

- The QL-circular geometry ( $\hat{k} \parallel \hat{B}$ ) is achievable only by transmitting toward magnetic SOUTH at elevation equal to the dip (down toward the dip pole, looking up at elevation = dip). At FC that's mag-S at 66 elevation – a real launch direction, but it gives 50/50 mode excitation for any linear antenna and so is useless for mode-discrimination purposes. Useful only as a *calibration* geometry where O and X are known to be excited equally.
- Pure single-mode launch in arbitrary directions requires electronic polarization control at the transmitter (e.g., crossed-dipole feeds with adjustable amplitude ratio). Not required if the receiver does the mode separation in software – see the orthogonal-magloop architecture documented elsewhere in this repository.

## Reproducibility

Source MATLAB / Python scripts and CSV / KML data products live in `~/Documents/MATLAB/pharlap/` (PHaRLAP work) and `~/Desktop/projects/ionosphere/kiwiclient/` (supporting tools). Specifically:

- `pharlap_qt_locus.m` – FC QT-locus enumeration with per-frequency map and KML export.
- `pharlap_qt_locus_taipei.m` – Taipei equivalent.
- `qt_polar_map.py` – the Stokes-style polar map, pure-Python (no PHaRLAP dependency).
- `build_qt_report_tables.py` – regenerate the LaTeX table fragments from the CSV outputs.

To regenerate this report:

```
# 1. Run the MATLAB scripts to (re)generate the CSVs and PNGs in
#    ~/Documents/MATLAB/pharlap/batch_out/
# 2. Then:
cd hf-polarimetry-release/reports/qt_locus
python3 ~/Documents/MATLAB/pharlap/build_qt_report_tables.py
pdflatex qt_locus_report.tex
pdflatex qt_locus_report.tex    # second pass for cross-refs
```

## References

- Cervera, M. A., *PHaRLAP – a HF radio propagation ray-tracing toolkit*, Defence Sci. & Tech. Group, Australia.
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